

MPBCDC Employee Management System — Detailed Implementation Guide

Purpose: Build a focused HRMS for MPBCDC covering **Loan Management**, **Leave Management**, and **Employee Service Records** — not a full payroll/attendance/compliance suite.

Source documents: HR_TECHNICAL_DOCUMENT_DETAILED.md (client field spec), MPBCDC Employee Management System.md (functional scope).

Reference architecture: Existing TalentCo HRMS (frontend-admin, frontend-client, frontend-employee, backend).

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1. Technology Stack

Layer	Technology	File Extensions	Notes
Frontend — Super Admin	React 18 + Vite	.jsx, .js	frontend-admin/ — tenant provisioning only
Frontend — Client	React 18 + Vite	.jsx, .js	frontend-

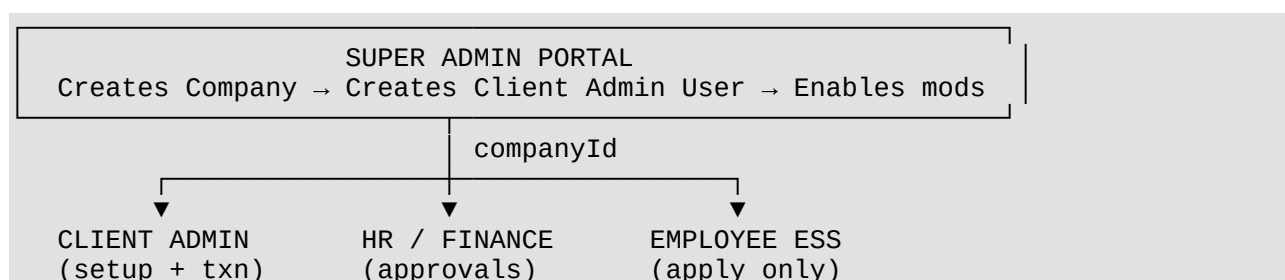
Layer	Technology	File Extensions	Notes
Admin			client/ — HR, Finance, Manager, Org Admin
Frontend — Employee ESS	React 18 + Vite	.jsx, .js	frontend-employee/ — self-service only
Backend API	Node.js + Express	.js	backend/ — REST API
Database	MongoDB + Mongoose	—	All tenant data scoped by companyId
Auth	JWT (Bearer token)	—	Role array in token payload
Charts	Chart.js	—	Dashboard widgets
File Upload	Multer → local/S3	—	PDF only for loan/leave attachments

Explicit rule for devs: All React UI components use `.jsx`. Utilities, API clients, middleware, controllers, models, and routes use `.js`.

2. Architecture & Multi-Tenancy

2.1 Tenant Model

Each MPBCDC deployment is one **Company (tenant)**. All business collections carry `companyId`.



2.2 Multi-Tenancy — Backend

Concern	Implementation	File
Tenant resolution	JWT companyId → <code>req.companyId</code> ; reject if missing	<code>backend/middleware/tenantResolver.js</code>
Data isolation	Every Mongoose query filters <code>{ companyId: req.companyId }</code>	All controllers

Concern	Implementation	File
Cross-tenant block	Strip <code>companyId</code> from request body	<code>tenantResolver.js</code>
Super Admin override	<code>SUPER_ADMIN</code> may pass <code>x-company-id</code> header to act on a tenant	<code>tenantResolver.js</code>
Indexes	Compound indexes on <code>{ companyId, ... }</code> on all collections	Model files

Expected result: Employee of Company A cannot read/write Company B data even with a valid token from Company A.

2.3 Multi-Tenancy — Frontend

Portal	LocalStorage namespace	Token key	Company context
Super Admin	<code>hrms_super_admin</code>	<code>auth_token</code>	Optional — selects tenant via header
Client Admin	<code>hrms_client_admin</code>	<code>hrms_client_admin:token</code>	From JWT <code>companyId</code>
Employee ESS	<code>hrms_employee</code>	<code>hrms_employee:token</code>	From JWT <code>companyId</code>

Frontend enforcement (mandatory):

1. `PrivateRoutes.jsx` in each app checks token **and** allowed roles before rendering `<Outlet />`.
2. Sidebar/menu built from role → module map (do not rely on hiding links alone).
3. `SubModuleGuard` (or equivalent) on sensitive routes — redirect to dashboard if role lacks permission.
4. Axios interceptor attaches `Authorization: Bearer <token>`; never send `companyId` in body.

Expected result: `CLIENT_ADMIN` user opening Employee portal URL is redirected to login with access-denied message. Employee cannot navigate to `/settings/*` even by typing URL.

3. Roles, Portals & Route Segregation

3.1 Role Matrix

Role Code	Portal	Primary Responsibilities
SUPER_ADMIN	frontend-admin	Create companies, global config, cross-tenant reports
CLIENT_ADMIN	frontend-client	All setup masters, users, workflow, notifications
HR_OFFICER	frontend-client	Employee master, service book, leave/loan HR approval
FINANCE_OFFICER	frontend-client	Loan disbursement, EMI recovery, closure, vouchers
REPORTING_MANAGER	frontend-client	Approve leave/loan for direct reports, view team service records
REGIONAL_MANAGER	frontend-client	Regional approvals (if matrix requires), regional reports
EMPLOYEE	frontend-employee	Apply loan/leave, view own balances, own documents

MPBCDC doc lists "Administrator" — map to CLIENT_ADMIN.

3.2 Login & Route Map

Super Admin Portal (frontend-admin)

Route	Page	Role
/login	Login	Public
/dashboard	Platform dashboard	SUPER_ADMIN
/companies	Company list/create	SUPER_ADMIN
/companies/:id	Company detail & module flags	SUPER_ADMIN
/companies/:id/users	Client admin user provisioning	SUPER_ADMIN
/audit-logs	Cross-tenant audit (read-only)	SUPER_ADMIN

Login URL example: `https://admin.mpbcdc.example.com/login`

Client Admin Portal (frontend-client)

Route prefix	Module	Roles
/login	Auth	Public
/dashboard	MIS dashboard	All client roles
/settings/	Dept, Designation, Grade,	CLIENT_ADMIN

Route prefix	Module	Roles
organization/*	Region, District	
/settings/employees/*	Employee master CRUD	CLIENT_ADMIN, HR_OFFICER
/settings/loan/*	Loan type, eligibility, workflow	CLIENT_ADMIN
/settings/leave/*	Leave type, accrual, holiday, workflow	CLIENT_ADMIN
/settings/workflow/*	Approval matrix	CLIENT_ADMIN
/settings/notifications/*	Notification templates	CLIENT_ADMIN
/settings/users/*	User & role assignment	CLIENT_ADMIN
/transactions/loans/*	Loan applications, approvals, disbursement	HR, Finance, Manager
/transactions/leaves/*	Leave applications, approvals	HR, Manager
/service-records/*	Employee service book	HR, Manager, CLIENT_ADMIN
/reports/*	All reports	Role-filtered

Login URL example: <https://portal.mpbcdc.example.com/login>

Employee ESS Portal (frontend-employee)

Route	Page	Role
/login	Login (Employee Code + Password)	Public
/dashboard	Personal dashboard	EMPLOYEE
/loans/apply	New loan application	EMPLOYEE
/loans/applied	My loan list & status	EMPLOYEE
/loans/:id/schedule	EMI repayment schedule	EMPLOYEE
/leaves/apply	New leave application	EMPLOYEE
/leaves/history	Leave history	EMPLOYEE
/leaves/balance	Leave balance	EMPLOYEE
/profile	Limited personal view	EMPLOYEE
/documents	Own documents only	EMPLOYEE
/notifications	In-app notifications	EMPLOYEE

Login URL example: <https://ess.mpbcdc.example.com/login>

3.3 Backend Route Authorization Pattern

Every API route chain:

```
router.use(auth); // JWT verify
router.use(tenantResolver); // companyId on req
router.get('/...', authorize('HR_OFFICER', 'CLIENT_ADMIN'), controller);
```

Mirror existing files: `authMiddleware.js`, `tenantResolver.js`, `authorizeRoles.js`.

Expected result: POST `/api/loans` with EMPLOYEE token creates application for self only. Same endpoint with EMPLOYEE token cannot create for another `employeeId`.

4. Module Pattern: Setup vs Transactions

Follow the existing HRMS convention — **Setup** configures rules; **Transactions** execute business flow.

```
MODULE
├── SETUP (Master Data)      → /settings/<module>/*
│   ├── Master CRUD
│   ├── Rules / Configuration
│   └── Workflow / Approval Matrix
└── TRANSACTIONS            → /transactions/<module>/*
    ├── Employee-initiated (apply)
    ├── Approver queues
    └── Post-approval processing (disbursement, balance update)
```

Module	Setup Routes (Client Admin)	Transaction Routes
Loan	/settings/loan/types, /settings/loan/eligibility, /settings/loan/workflow	/transactions/loans/applications, /transactions/loans/disbursement, /transactions/loans/recovery
Leave	/settings/leave/types, /settings/leave/accrual, /settings/leave/holidays, /settings/leave/workflow	/transactions/leaves/applications, /transactions/leaves/approvals, /transactions/leaves/balances
Employee	/settings/organization/*, /settings/employees	Service record transactions under /service-records/*

5. Employee Master (Scoped Fields Only)

Do not use the full HRMS Employee model (400+ fields). Create a slim MpbcdcEmployee model (or scoped schema) with **only** client-required fields.

5.1 Field Mapping

Client Field	DB Field	Type	Mandatory	UI Control	Notes
Employee ID	_id	ObjectId	Auto	Hidden	MongoDB primary key
Employee Code	employeeCode	String(20)	Yes	Text	Unique per companyId
Employee Name	employeeName	String(200)	Yes	Text	Single field (not first/last split)
Gender	gender	String(20)	Yes	Dropdown	Male / Female / Other
Date of Birth	dateOfBirth	Date	Yes	Date picker	
Joining Date	joiningDate	Date	Yes	Date picker	Used for service period & leave pro-rata
Retirement Date	retirementDate	Date	Yes	Date picker	Used for loan retirement buffer rule
Mobile Number	mobileNumber	String(15)	Yes	Text	10-digit validation
Email ID	email	String(100)	Yes	Email	Login optional secondary
Aadhaar Number	aadhaarNumber	String(12)	Yes	Text	Mask display in UI
PAN Number	panNumber	String(10)	Yes	Text	Uppercase
Department	departmentId	ObjectId	Yes	Lookup	→ Department
Designation	designationId	ObjectId	Yes	Lookup	→ Designation
Grade	gradeId	ObjectId	Yes	Lookup	→ Grade
Region	regionId	ObjectId	Yes	Lookup	→ Region
District	districtId	ObjectId	Yes	Lookup	→ District
Reporting Manager	reportingManagerId	ObjectId	Yes	Employee lookup	Same company, cannot be self

Client Field	DB Field	Type	Mandatory	UI Control	Notes
Employment Type	employment Type	String(50)	Yes	Dropdown	Permanent / Contract / etc.
Status	status	String(20)	Yes	Toggle	Active / Inactive
—	companyId	ObjectId	Yes	Hidden	Tenant scope
—	grossSalary	Number	Yes*	Number	*Required for loan eligibility; HR maintains via salary revision txn

5.2 Employee Master — By Layer

Layer	Responsibility	Expected Result
DB	Collection mpbcdc_employees; index { companyId, employeeCode } unique	No duplicate codes within tenant
Backend	POST/GET/PUT /api/employees; validate lookups belong to same companyId; block delete if open loans/leaves	Inactive employee cannot log in
Frontend (Setup)	frontend-client/src/pages/settings/employees/EmployeeList.jsx, EmployeeForm.jsx	HR/Admin sees paginated list, create/edit with lookup dropdowns
Frontend (ESS)	frontend-employee/src/pages/Profile.jsx — read-only subset	Employee sees name, dept, designation, grade, manager, joining date only

6. Organization Setup Masters

All masters: companyId scoped, status: Active|Inactive, effective date where applicable.

6.1 Department Master

Field	DB Field	Type
Department ID	_id	ObjectId
Department Name	name	String

Field	DB Field	Type
Department Head	headEmployeeId	ObjectId → Employee
Effective Date	effectiveDate	Date
Status	status	Active/Inactive

Layer	Files / Endpoints
DB	models/Department.js
Backend	routes/departmentRoutes.js, controllers/departmentController.js — GET/POST/PUT /api/departments
Frontend	pages/settings/organization/ DepartmentMaster.jsx

6.2 Designation Master

Field	DB Field
Designation ID	_id
Designation Name	name
Grade	gradeId (lookup)
Pay Scale	payScale (String)
Status	status

6.3 Grade Master

Field	DB Field
Grade Code	code
Grade Name	name
Approval Matrix Applicable	approvalMatrixApplicable (Boolean)
Status	status

6.4 Region Master

Field	DB Field
Region Code	code
Region Name	name
Regional Manager	managerEmployeeId

6.5 District Master

Field	DB Field
District Code	code
District Name	name
Region Mapping	regionId

Expected result: Dropdowns in Employee form cascade — District list filters by selected Region.

7. Loan Management Module

7.1 Module Overview

LOAN MODULE
└─ SETUP
└─ Loan Type Master
└─ Loan Eligibility Configuration
└─ Approval Workflow (via Workflow Master)
└─ TRANSACTIONS
└─ Loan Application (Employee)
└─ Loan Approval (Manager → HR → Finance)
└─ Loan Disbursement (Finance)
└─ EMI Schedule (auto-generated)
└─ Loan Recovery (payroll month)
└─ Loan Closure

7.2 LOAN — Master Data Setup

7.2.1 Loan Type Master

Client Field	DB Field	Type	Notes
Loan Type Code	code	String	e.g. EDU, COMP, VEH, HOME, MARR
Loan Type Name	name	String	Education, Computer, Vehicle, Home, Marriage
Education Loan	isEducationLoan	Boolean	Flags for reporting
Computer Loan	isComputerLoan	Boolean	
Vehicle Loan	isVehicleLoan	Boolean	
Home Loan	isHomeLoan	Boolean	
Marriage Loan	isMarriageLoan	Boolean	
Maximum Amount	maxAmount	Number	
Maximum Tenure	maxTenureMonths	Number	
Interest Rate	interestRate	Number	Annual %

Client Field	DB Field	Type	Notes
Minimum Service Years	minServiceYears	Number	
Salary Multiplier	salaryMultiplier	Number	e.g. 10 = 10× gross
Active Flag	isActive	Boolean	

Layer	Implementation
DB	models/LoanType.js — { companyId, code } unique
Backend	routes/loanTypeRoutes.js — CRUD /api/loan-types — roles: CLIENT_ADMIN
Frontend	pages/settings/loan/LoanTypeList.jsx, LoanTypeForm.jsx

Expected result: Admin creates "Education Loan" with max ₹5,00,000, 60 months, 8% interest. Employee apply screen shows only active types.

7.2.2 Loan Eligibility Configuration Master

Client Field	DB Field	Default
Eligibility Rule ID	ruleCode	DEFAULT
Minimum Service Period	minServiceMonths	Per loan type or global
Salary Multiplier	salaryMultiplier	From loan type
Maximum EMI Percentage	maxEmiPercentOfGross	60
Retirement Buffer Months	retirementBufferMonths	3
Effective Date	effectiveDate	
Status	status	Active

Default business rules (backend service):

1. $EMI \leq 60\%$ of gross salary (after existing EMI).
2. Employee retains $\geq 40\%$ gross after all EMIs.
3. Loan must close ≥ 3 months before `retirementDate`.
4. Existing active loan EMIs summed automatically.

Layer	Implementation
DB	models/LoanEligibilityRule.js

Layer	Implementation
Backend	services/ loanEligibilityService.js — pure function calculateEligibility(employee, loanType, requestedAmount, requestedTenure)
Frontend	pages/settings/loan/ LoanEligibilityConfig.jsx

Expected result: On apply form, when employee enters amount/tenure, UI calls GET /api/loans/preview-eligibility and shows derived fields before submit.

7.3 LOAN — Approval Workflow

Workflow States

Draft → Submitted → Manager Approval → HR Approval → Finance Approval →
Disbursed → Closed
 ↳ Rejected (terminal)

Approval Matrix (Setup)

Uses shared **Workflow Master** (Section 10). For loans:

Level	Approver Role	Escalation Days
1	REPORTING_MANAGER	3
2	HR_OFFICER	3
3	FINANCE_OFFICER	5

Layer	Implementation
DB	models/LoanApproval.js — one row per approval action
Backend	services/loanWorkflowService.js — getNextApprover(), canApprove(user, application)
Frontend	pages/settings/loan/ LoanApprovalWorkflow.jsx (setup); pages/transactions/loans/LoanApp rovalQueue.jsx (txn)

Approval record fields:

Field	DB Field
Application No	applicationId
Approver Role	approverRole
Approver Name	approverUserId (populate)
Approval Date	approvedAt
Decision	decision — Approved / Rejected
Remarks	remarks
Digital Signature	signatureFilePath or signatureDataUrl
Next Approver	nextApproverRole

Expected result: HR approval button disabled until Manager approves. Finance queue shows only HR-approved applications. Rejection at any level sets status **Rejected** and notifies employee.

7.4 LOAN — Transactions

7.4.1 Loan Application

Header fields (employee input):

Field	DB Field	Type
Application No	applicationNo	Auto LN-YYYY-#####
Application Date	applicationDate	Date
Employee ID	employeeId	ObjectId
Loan Type	loanTypeId	ObjectId
Requested Amount	requestedAmount	Decimal
Requested Tenure	requestedTenureMonths	Integer
Purpose	purpose	Text
Covering Note	coveringNote	Text
Attachment	attachmentPath	PDF

System-derived fields (stored on submit):

Field	DB Field
Gross Salary	grossSalaryAtApplication
Existing EMI	existingEmiTotal
Maximum Eligible Amount	maxEligibleAmount
Maximum Eligible EMI	maxEligibleEmi
Remaining Service Years	remainingServiceYears

Field	DB Field
Eligibility Status	eligibilityStatus — Eligible / Ineligible
Ineligibility Reasons	ineligibilityReasons[]

Layer	Implementation
DB	models/LoanApplication.js
Backend	POST /api/loan-applications (EMPLOYEE); PUT /:id/submit; block submit if eligibilityStatus !== 'Eligible'
Frontend (ESS)	pages/loans/ApplyLoan.jsx — wizard: select type → amount/tenure → eligibility panel → upload → submit
Frontend (Admin)	pages/transactions/loans/LoanApplicationList.jsx — filters by status, role-based queues

Expected result: Employee with 1 year service applying for Home Loan with 5-year min service sees "Ineligible: Service period insufficient" and Submit is disabled.

7.4.2 Loan Disbursement

Field	DB Field
Disbursement Number	disbursementNo
Loan Number	loanNo — assigned at disbursement
Sanction Amount	sanctionAmount
Interest Rate	interestRate
EMI Amount	emiAmount
Disbursement Date	disbursementDate
Bank Account	bankAccount
Finance Remarks	financeRemarks
Voucher Number	voucherNo

Layer	Implementation
DB	models/LoanDisbursement.js
Backend	POST /api/loan-applications/:id/disburse — FINANCE_OFFICER only; triggers EMI schedule generation
Frontend	pages/transactions/loans/

Layer	Implementation
	LoanDisbursementForm.jsx

Business rule: If disbursement date is after 20th of month → first EMI due next month.

Expected result: After disbursement, application status = **Disbursed**, employee receives notification, EMI schedule visible in ESS.

7.4.3 EMI Schedule (auto-generated)

Field	DB Field
Schedule ID	_id
Loan Number	loanNo
EMI No	emiNo
Due Date	dueDate
Principal	principalComponent
Interest	interestComponent
EMI Amount	emiAmount
Outstanding Balance	outstandingBalance
Deduction Status	deductionStatus — Pending / Deducted / Skipped

Layer	Implementation
DB	models/LoanEmiSchedule.js
Backend	services/loanScheduleService.js — generateSchedule(loanNo, sanctionAmount, rate, tenure, startDate) — reducing balance
Frontend (ESS)	pages/loans/ RepaymentSchedule.jsx — table + progress bar
Frontend (Admin)	pages/transactions/loans/ EmiScheduleView.jsx

7.4.4 Loan Recovery

Field	DB Field
Recovery ID	_id
Loan Number	loanNo
Payroll Month	payrollMonth — YYYY-MM
EMI Amount	emiAmount

Field	DB Field
Recovery Date	recoveryDate
Recovery Status	status
Balance Outstanding	balanceOutstanding

Layer	Implementation
DB	models/LoanRecovery.js
Backend	POST /api/loan-recoveries — mark EMI deducted for payroll month; update schedule row
Frontend	pages/transactions/loans/LoanRecoveryRegister.jsx

7.4.5 Loan Closure

Field	DB Field
Closure Number	closureNo
Loan Number	loanNo
Closure Date	closureDate
Outstanding Amount	outstandingAmount
Closure Type	closureType — Regular / Prepayment / Settlement
Closure Certificate No	certificateNo

Expected result: Closed loan no longer appears in active EMI deduction. Employee can download closure certificate PDF.

8. Leave Management Module

8.1 Module Overview

LEAVE MODULE	
— SETUP	
—	Leave Type Master
—	Holiday Master
—	Leave Accrual Rule Master
—	Approval Workflow
— TRANSACTIONS	
—	Leave Balance (ledger)
—	Leave Application
—	Leave Approval → Balance Update

Supported leave types: CL (Casual), SL (Sick), EL (Earned), Special Leave.

8.2 LEAVE — Master Data Setup

8.2.1 Leave Type Master

Client Field	DB Field	Type
Leave Code	code	String — CL, SL, EL, SPL
Leave Name	name	String
Annual Entitlement	annualEntitlement	Number
Half Day Allowed	halfDayAllowed	Boolean
Encashment Allowed	encashmentAllowed	Boolean
Carry Forward Allowed	carryForwardAllowed	Boolean
Maximum Accumulation	maxAccumulation	Number — EL default 300
Sandwich Rule Applicable	sandwichRuleApplicable	Boolean
HR Approval Required	hrApprovalRequired	Boolean

Layer	Implementation
DB	models/LeaveType.js
Backend	/api/leave-types CRUD
Frontend	pages/settings/leave/LeaveTypeMaster.jsx

8.2.2 Holiday Master

Field	DB Field
Holiday ID	_id
Holiday Name	name
Holiday Type	type — National / Regional / Optional
Holiday Date	date
State	state
Applicable Region	regionId

Layer	Implementation
DB	models/Holiday.js
Backend	/api/holidays
Frontend	pages/settings/leave/HolidayCalendar.jsx

8.2.3 Leave Accrual Rule Master

Field	DB Field	EL Default
Rule ID	ruleCode	
Leave Type	leaveTypeId	EL
Frequency	frequency	Half-Yearly
Credit Days	creditDays	15
Pro-rata Applicable	proRataApplicable	true
Accumulation Limit	accumulationLimit	300
Effective Date	effectiveDate	

Layer	Implementation
DB	models/LeaveAccrualRule.js
Backend	services/leaveAccrualService.js — cron/job on Jan 1 & Jul 1; pro-rata for new joiners
Frontend	pages/settings/leave/LeaveAccrualSetup.jsx

Expected result: New joiner on April 1 receives pro-rated EL credit. Accrual job idempotent — running twice does not double-credit.

8.3 LEAVE — Approval Workflow

Employee → Reporting Manager → HR (optional per leave type) → Balance Update → Notification

Layer	Implementation
DB	models/LeaveApproval.js
Backend	services/leaveWorkflowService.js
Frontend (Setup)	pages/settings/leave/LeaveApprovalWorkflow.jsx
Frontend (Txn)	pages/transactions/leaves/LeaveApprovalQueue.jsx

Approval fields: Application No, Approver, Approval Date, Decision, Remarks, Escalation Status.

Expected result: Manager rejects → balance not deducted. Manager approves + HR required → waits for HR. All approved → `leaveBalanceService.deduct()` runs atomically.

8.4 LEAVE — Transactions

8.4.1 Leave Balance (Ledger)

Field	DB Field
Employee ID	employeeId
Leave Type	leaveTypeId
Opening Balance	openingBalance
Accrued	accrued
Availed	availed
Closing Balance	closingBalance
Period	period — e.g. 2026 or 2026-H1

Layer	Implementation
DB	models/LeaveBalance.js — index { companyId, employeeId, leaveTypeId, period }
Backend	GET /api/leave-balances/my (EMPLOYEE); GET /api/leave-balances (HR)
Frontend (ESS)	pages/leaves/LeaveBalance.jsx
Frontend (Admin)	pages/transactions/leaves/LeaveBalanceReport.jsx

8.4.2 Leave Application

Field	DB Field
Leave Application No	applicationNo
Employee ID	employeeId
Leave Type	leaveTypeId
From Date	fromDate
To Date	toDate
Total Days	totalDays — system calculated
Half Day Flag	isHalfDay
Reason	reason
Attachment	attachmentPath
Current Balance	balanceBefore
Balance After Approval	balanceAfter
Status	status — Draft / Submitted / Approved / Rejected

Layer	Implementation
DB	models/LeaveApplication.js
Backend	services/ leaveCalculationService.js — working days, holidays, sandwich rule
Frontend (ESS)	pages/leaves/ApplyLeave.jsx

Sandwich rule logic (when enabled on leave type):

Fri leave + Sat-Sun holiday + Mon leave = 4 days deducted (not 2)

Layer	Sandwich Implementation
Backend	leaveCalculationService.calculateDays(from, to, employee.regionId, leaveType)
Frontend	Show breakdown: Working days / Holidays sandwiched / Total deducted

Expected result: Employee with 2 CL balance applying for 3 days sees validation error before submit. Approved leave reduces balance; rejected does not.

9. Employee Service Record Module

Visible to HR, CLIENT_ADMIN, and REPORTING_MANAGER (direct reports only).
Not visible to EMPLOYEE (except limited profile).

9.1 Structure

SERVICE RECORD

— Header (read-only snapshot)	
— Promotion History	(transaction)
— Salary Revision	(transaction)
— Transfer History	(transaction)
— Disciplinary Action	(transaction)
— Document Repository	(transaction)

9.1.1 Service Record Header

Field	Source
Service Record Number	Auto SR-YYYY-#####
Employee ID	employeeId
Employee Name	from Employee
DOJ	joiningDate

Field	Source
Current Designation / Grade / Department / Region	live from Employee

9.2 Transactions — By Layer

Transaction	DB Model	Backend Route	Frontend Page
Promotion History	PromotionHistory.js	POST /api/service-records/:employeeId/promotions	ServiceRecord/PromotionForm.jsx
Salary Revision	SalaryRevision.js	POST .../salary-revisions	SalaryRevisionForm.jsx — updates employee.grossSalary
Transfer History	TransferHistory.js	POST .../transfers	TransferForm.jsx — updates region/dept on employee
Disciplinary Action	DisciplinaryCase.js	POST .../disciplinary	DisciplinaryForm.jsx
Document Repository	EmployeeDocument.js	POST .../documents (multipart)	DocumentUpload.jsx

Document categories: Appointment Letter, Promotion Order, Transfer Order, Educational Certificates, Identity Documents, Disciplinary Documents, Retirement Documents.

Expected result: Manager opens service record for reportee — sees promotion/transfer history. Manager cannot open unrelated employee. Salary revision reflects in next loan eligibility calculation.

10. Workflow, Notifications & Audit

10.1 Workflow Master (Shared Setup)

Field	DB Field
Workflow Code	code
Module	module — LOAN / LEAVE
Grade	gradeId
Level	level — 1, 2, 3...
Approver Role	approverRole

Field	DB Field
Escalation Days	escalationDays
Active	isActive

Layer	Files
DB	models/ApprovalMatrix.js
Backend	/api/workflow-matrix
Frontend	pages/settings/workflow/ApprovalMatrix.jsx

10.2 Notification Master (Setup)

Field	DB Field
Template ID	templateCode
Event Name	eventName
Email Template	emailBody
SMS Template	smsBody
WhatsApp Template	whatsappBody
Trigger Stage	triggerStage

Events: Loan Submitted, Loan Approved, Loan Rejected, Leave Approved, Leave Rejected, Promotion Updated, EMI Reminder, Retirement Reminder.

Layer	Implementation
DB	models/NotificationTemplate.js, models/Notification.js (in-app inbox)
Backend	services/notificationService.js — called from workflow hooks
Frontend	pages/settings/notifications/NotificationTemplates.jsx; bell icon inbox all portals

10.3 User & Role Master

Field	DB Field
User ID	_id
Login ID	loginId — Employee Code
Password Hash	passwordHash
Employee ID	employeeId

Field	DB Field
Role	roles[]
Company ID	companyId
Last Login	lastLoginAt

Layer	Implementation
DB	models/User.js
Backend	/api/auth/login, /api/users
Frontend	pages/settings/users/UserManagement.jsx

10.4 Audit

Layer	Implementation
DB	models/AuditLog.js — { companyId, userId, action, entity, entityId, before, after, timestamp }
Backend	Mongoose plugin or middleware on write operations
Frontend	pages/settings/audit/AuditLogList.jsx — CLIENT_ADMIN only

11. Reports & Dashboards

11.1 Loan Reports

Report	Endpoint	Roles
Loan Eligibility Report	GET /api/reports/loans/eligibility	HR, Admin
Loan Application Register	GET /api/reports/loans/applications	HR, Finance, Admin
Loan Sanction Register	GET /api/reports/loans/sanctions	Finance, Admin
Disbursement Report	GET /api/reports/loans/disbursements	Finance

Report	Endpoint	Roles
EMI Recovery Register	GET /api/reports/loans/recoveries	Finance
Outstanding Loan Report	GET /api/reports/loans/outstanding	Finance, HR
Loan Closure Report	GET /api/reports/loans/closures	Finance

11.2 Leave Reports

Report	Endpoint
Employee Leave Balance	GET /api/reports/leaves/balances
Leave Register	GET /api/reports/leaves/register
Pending Approvals	GET /api/reports/leaves/pending
Leave Utilization	GET /api/reports/leaves/utilization
Half Day Leave	GET /api/reports/leaves/half-day

11.3 MIS Dashboard Cards

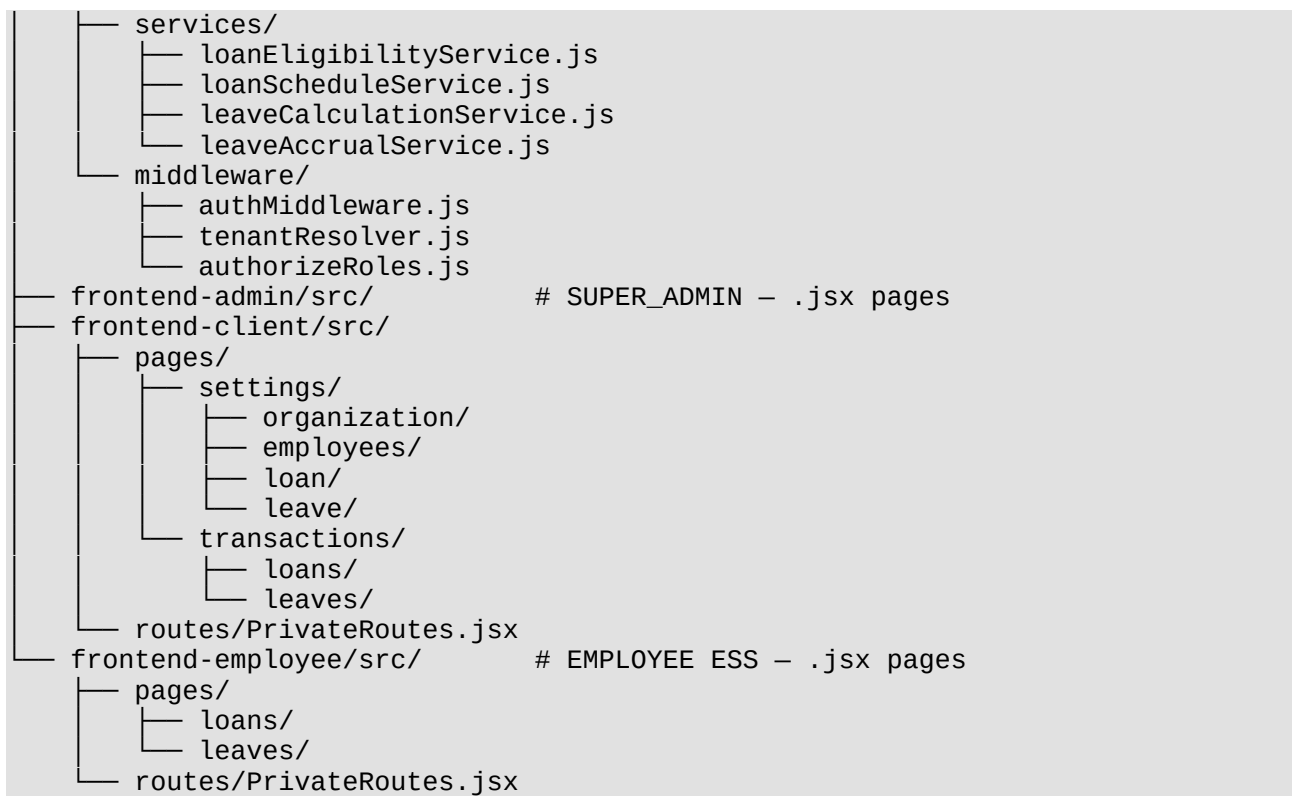
Portal	Widgets
Client Admin	Total/Active Employees, Pending Loans, Pending Leaves, Outstanding Loans, Retirement Due
Employee	Leave balances, Active loan EMI, Next EMI date, Pending requests

12. Folder Structure (Recommended)

```

hrms-app/
├── backend/
│   ├── models/
│   │   ├── MpbcdcEmployee.js
│   │   ├── Department.js
│   │   ├── LoanType.js
│   │   ├── LoanApplication.js
│   │   ├── LoanEmiSchedule.js
│   │   ├── LeaveType.js
│   │   ├── LeaveApplication.js
│   │   └── ...
│   ├── controllers/
│   └── routes/

```

13. API Conventions

Convention	Rule
Base path	/api
Response shape	{ success: true, data: ... } or { success: false, error: '...' }
Pagination	?page=1&limit=20
Auto numbers	LN-2026-00001, LV-2026-00001, SR-2026-00001 — per company sequence
File upload	multipart/form-data; max 5MB; PDF only
Dates	ISO 8601 in API; display DD-MMM-YYYY in UI

14. Implementation Phases

Phase 1 — Foundation (Week 1–2)

- ☐ Company + User + Auth + tenant middleware
- ☐ Three frontends with PrivateRoutes role gates
- ☐ Organization masters + slim Employee master

Done when: Super Admin creates company; Client Admin logs in and creates employee; Employee logs in to ESS.

Phase 2 — Leave Module (Week 3–4)

- ☐ Leave setup (types, holidays, accrual, workflow)
- ☐ Leave application + approval + balance ledger
- ☐ Sandwich rule + half-day

Done when: End-to-end leave apply → manager approve → balance reduced.

Phase 3 — Loan Module (Week 5–7)

- ☐ Loan setup (types, eligibility)
- ☐ Application + multi-level approval
- ☐ Disbursement + EMI schedule + recovery + closure

Done when: End-to-end loan with eligibility block, 3-level approval, EMI schedule visible in ESS.

Phase 4 — Service Record + Reports (Week 8–9)

- ☐ Service book transactions + documents
- ☐ Reports + dashboards
- ☐ Notifications + audit

Done when: HR maintains service book; all client-listed reports exportable.

15. Acceptance Criteria Summary

#	Scenario	Expected Result
1	Employee opens frontend-client URL	Redirected to login; access denied
2	Client Admin creates employee	Employee code unique per company; lookups validated
3	Employee applies leave spanning sandwich weekend	Days deducted per sandwich rule when enabled
4	Employee applies loan exceeding EMI limit	Submit blocked; reasons displayed
5	Manager approves loan	HR queue shows application; Finance queue empty
6	Finance disburses on 25th	First EMI due next month
7	Leave approved	closingBalance reduced by totalDays
8	HR uploads promotion order	Visible in service record; not in

#	Scenario	Expected Result
		employee ESS
9	Two companies in same DB	Zero data leakage across companyId
10	Inactive employee login	Authentication fails

Appendix A — Role → Menu Visibility (Frontend)

Menu Item	CLIENT_A DMIN	HR	FINANCE	MANAGER	REGIONAL_MGR	EMPLOYEE
Organization Setup	✓	—	—	—	—	—
Employee Master	✓	✓	—	—	—	—
Loan Setup	✓	—	—	—	—	—
Leave Setup	✓	—	—	—	—	—
Loan Approvals	✓	✓	✓	✓	—	—
Loan Disbursement	—	—	✓	—	—	—
Leave Approvals	✓	✓	—	✓	—	—
Service Records	✓	✓	—	✓*	—	—
Apply Loan/Leave	—	—	—	—	—	✓
Reports	✓	✓	✓	✓	✓	—

*Manager: direct reports only.

Appendix B — Collections List (MongoDB)

companies
users
mpbcdc_employees
departments
designations
grades
regions
districts
loan_types

```
loan_eligibility_rules
loan_applications
loan_approvals
loan_disbursements
loan_emi_schedules
loan_recoveries
loan_closures
leave_types
holidays
leave_accrual_rules
leave_balances
leave_applications
leave_approvals
promotion_histories
salary_revisions
transfer_histories
disciplinary_cases
employee_documents
approval_matrices
notification_templates
notifications
audit_logs
```

Document version: 1.0 — Generated for MPBCDC implementation. Aligns with HR_TECHNICAL DOCUMENT_DETAILED.md field spec and TalentCo HRMS multi-portal architecture.